Week 5 Discussion

Covers: the probability integral transformation and expected values (Lecture 4.2); moments, variance, and mgfs (Lecture 5.1); what the mgf determines.

Problem 1 (The probability integral transformation)

Let X have pdf

$$f_X(x) = \begin{cases} \dfrac{x-1}{2} & 1 < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the cdf $F_X(x)$. Then find a monotone function $u(x)$ such that the random variable $Y = u(X)$ has a uniform(0, 1) distribution. Verify your answer directly, by computing $P(Y \le y)$ for $0 < y < 1$. *Hint:* Theorem 2.1.10.

(b) Now go the other way. Let $V \sim$ uniform(0, 1). Find a function h such that $h(V)$ has the same distribution as X, and check that $P(h(V) \le x) = F_X(x)$ for $1 < x < 3$. How would you use this to simulate X on a computer?

(c) Find $\mathrm{E}X$ in two ways:

- directly from f_X, by (2.2.4);
- as $\mathrm{E}\, h(V)$, using your answer to (b) and the uniform pdf.

Then use the fact that $\mathrm{E}F_X(X) = \frac{1}{2}$ to find $\mathrm{E}(X-1)^2$ **without** integrating against f_X, and deduce $\mathrm{Var}\, X$.

(d) Theorem 2.1.10 needs a *continuous* cdf. Let X be uniform on $\{1, 2, 3\}$, that is, $P(X = k) = \frac{1}{3}$ for $k = 1, 2, 3$, and let $Y = F_X(X)$.

- Find the pmf of Y and $\mathrm{E}Y$. Is Y uniform(0, 1)?
- Show that $F_Y(y) \le y$ for all $0 < y < 1$, with strict inequality for some y. (C&B say Y is *stochastically greater* than a uniform(0, 1).)
- Where does the proof of Theorem 2.1.10 use continuity of F_X, and why does that step fail here?

▶ *Show answer*

Problem 2 (MGF)

Let X have the geometric pmf

$$P(X = x) = p(1-p)^x, \qquad x = 0, 1, 2, \ldots, \quad 0 < p < 1.$$

X counts the failures before the first success in independent trials with success probability p. (Week 4 Discussion Problem 2 used $p = \frac{1}{3}$.)

(a) Verify that

$$M_X(t) = \frac{p}{1 - (1-p)e^t}.$$

For which t does the sum converge? Does the mgf exist in the sense of Definition 2.3.6?

(b) Use Theorem 2.3.7 to find $\mathrm{E}X$, $\mathrm{E}X^2$, and $\mathrm{Var}\, X$.

(c) (C&B Exercise 2.32) Let $S(t) = \log M_X(t)$, for any random variable X with an mgf. Show that

$$S'(0) = \mathrm{E}X \quad \text{and} \quad S''(0) = \mathrm{Var}\, X.$$

Then redo (b) for the geometric with $S(t)$. Which way is less work?

▶ *Show answer*

Problem 3 (MGF)

(a) Each function below is claimed to be the mgf of some random variable X. Identify the distribution of X, or explain why no such X exists. *Hint:* Theorems 2.3.11 and 2.3.15, and Examples 2.3.8 and 2.3.9.

- **(i)** $M(t) = \left(\frac{1}{3} + \frac{2}{3}e^t\right)^5$, for all t.
- **(ii)** $M(t) = \dfrac{e^{2t}}{1 - 3t}$, for $t < \frac{1}{3}$.
- **(iii)** $M(t) = \dfrac{t}{1 - t}$, for $|t| < 1$.

(b) For $n = 2, 3, \ldots$, let X_n have the geometric pmf of Problem 2 with $p = \frac{1}{n}$, and let $Y_n = X_n / n$. Think of trials run every $\frac{1}{n}$ seconds, each succeeding with probability $\frac{1}{n}$, so that Y_n is the waiting time, in seconds, before the first success.

- Use Problem 2(a) and Theorem 2.3.15 to find $M_{Y_n}(t)$.
- Show that $M_{Y_n}(t) \to \frac{1}{1-t}$ for every $t < 1$, and say what Theorem 2.3.12 lets you conclude about F_{Y_n}. *Hint:* $n\left(e^{t/n} - 1\right) \to t$.

▶ *Show answer*